

Scripting

- ⇒ Scripting should start with **#!/bin/bash**.
- ⇒ **#!/bin/bash** this line will be treated as shebang line from the system and it will know this file belongs to the bash script.
- ⇒ First we have to go nano editor to insert the text to the file

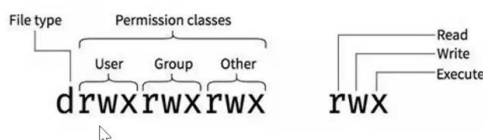
What is Script?

- **Definition:** A script is a plain text file with commands written in a shell language (commonly Bash in Linux).
- **Purpose:** Instead of typing commands one by one, you save them in a file and run the file to execute all commands in order.
- **File extension:** Often .sh, but not mandatory.

File Permissions.

FILE PERMISSION

- ☐ user – Permissions apply only to the owner of the file or directory, this will not impact the actions of other users.
- ☐ group – Group permissions apply only to the group that has been assigned to the file or directory, and will not effect the actions of other users.
- ☐ others – Others permissions apply to all other users on the system.



Owner		rwX	$4 + 2 + 1 = 7$
Group		rwX	$4 + 2 + 1 = 7$
Other		rw-	$4 + 2 = 6$

```
$chmod ugo+x abc.sh  
$chmod +x abc.sh  
$chmod 777 abc.sh
```

There were 3 types of permission or 3 entities for files and directories

Below will know file type and permission field if file starts with **drwxrwxrwx** and here d stand for the type of file which means directory and remaining 9 terms are permissions

- Owner – rwx(read write and execute)
- Group – rwx(read write and execute)
- Others - rwx(read write and execute)

If it starts with “-” for example **-rwxrwxrwx** here this – is for representing it as file not a directory

\$Chmod ugo+x abc.sh

Here ugo+x abc.sh means “u” stand for user, “g” stands for group, “o” stands for others and “+x” stands for executable permissions and the “chmod” commands confirms that giving the executable permissions for all of the users and here abc.sh is file name.

\$Chmod +x abc.sh

The above command also confirms that giving executable permission for all the files same as ***\$Chmod ugo+x abc.sh***

\$chmod 777 abc.sh

This command will confirm that all the permission for all the users.

➔ Chmod ug+x sample.sh – this will give executable permission for

To execute a script file we have to give ./before file name.

➔ Chmod ug-x sample.sh to remove the permissions for a file

➔ When we give executable permissions to a file it will change the colour of name of the file and when you remove it Vice versa.